

Final Business Insights

Treatment C is the most effective treatment for reducing blood pressure and improving patient recovery outcomes, while Treatment A is the least effective.

Best Treatment

Treatment C

- 100% recovery rate
- Highest BP reduction (41.67)
- Fastest recovery (5–6 days)
- Lowest pain score

Worst Treatment

Treatment A

- 0% recovery
- Lowest BP reduction
- Longest recovery duration

Healthcare Recommendation

If this dataset represents a clinical trial:

1. Promote Treatment C for larger-scale testing.
2. Investigate why Treatment A failed.
3. Collect more samples (9 records is too small for statistical significance).
4. Perform hypothesis testing (t-test/ANOVA) before making medical conclusions.

Conclusion

Treatment effectiveness ranking:

Treatment C > Treatment B > Treatment A

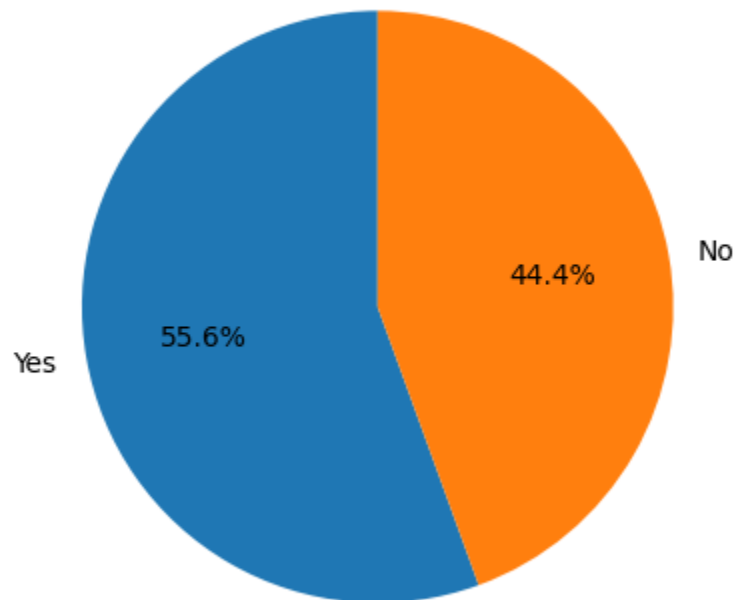
with clear improvements in:

- Recovery Rate
- Blood Pressure Reduction
- Recovery Days
- Pain Score

Recovery Distribution

Recovered vs not recovered patients.

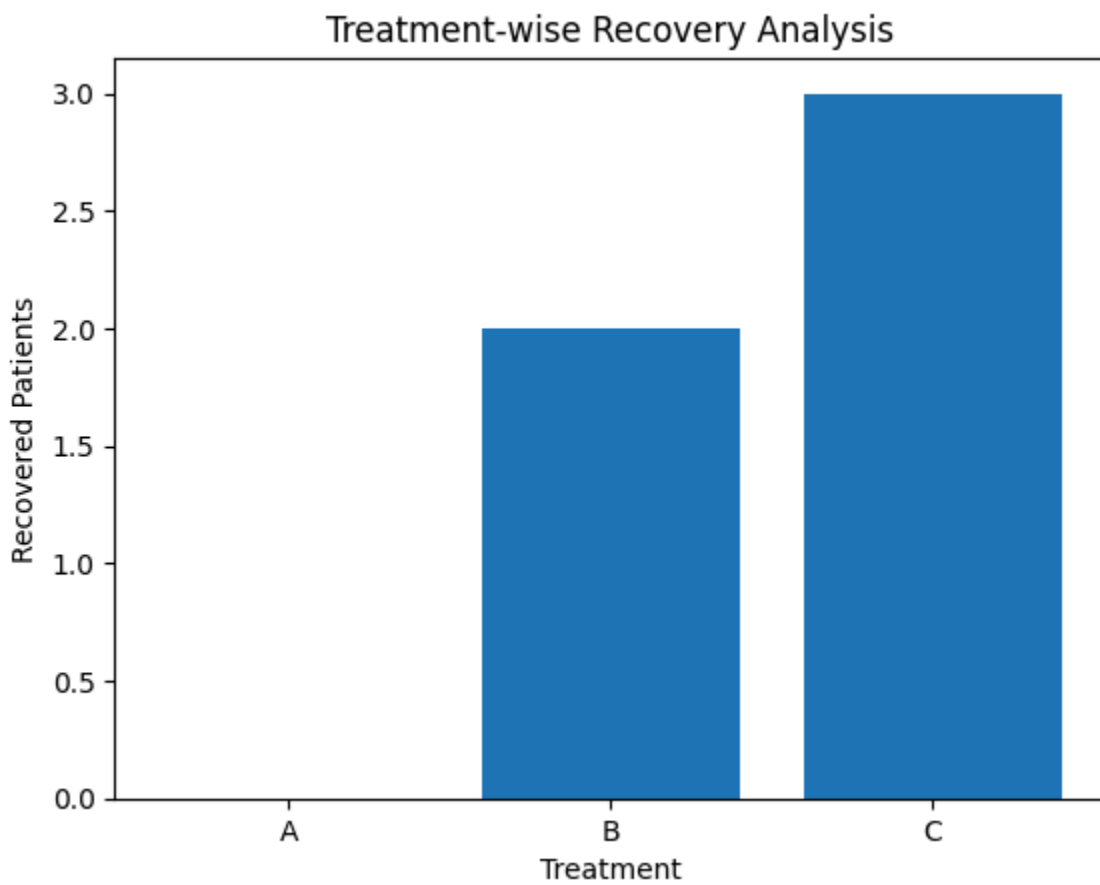
status	count
Recovered	5
Not Recovered	4



Recovery by Treatment

Number of recovered patients under each treatment.

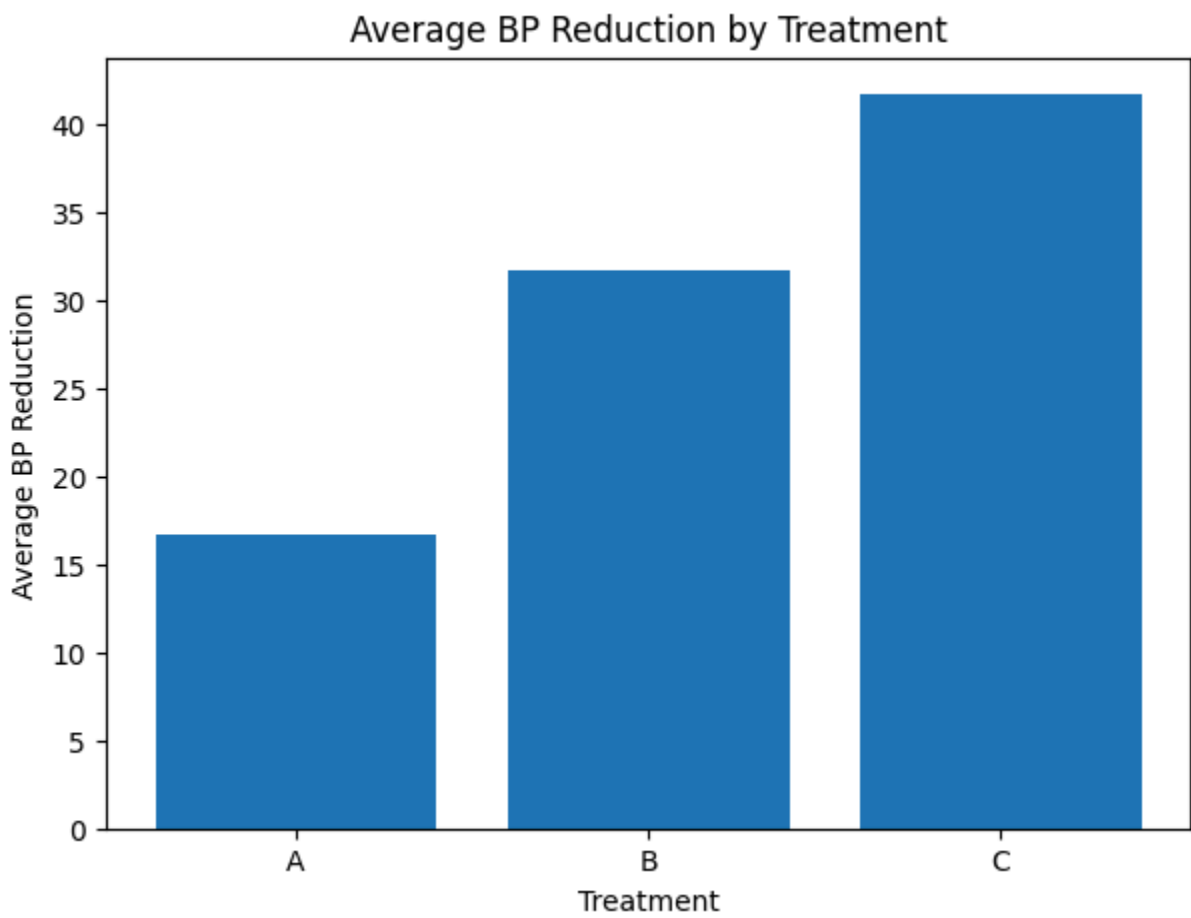
treatment	recovered
A	0
B	2
C	3



Average BP Reduction by Treatment

Comparison of treatment effectiveness.

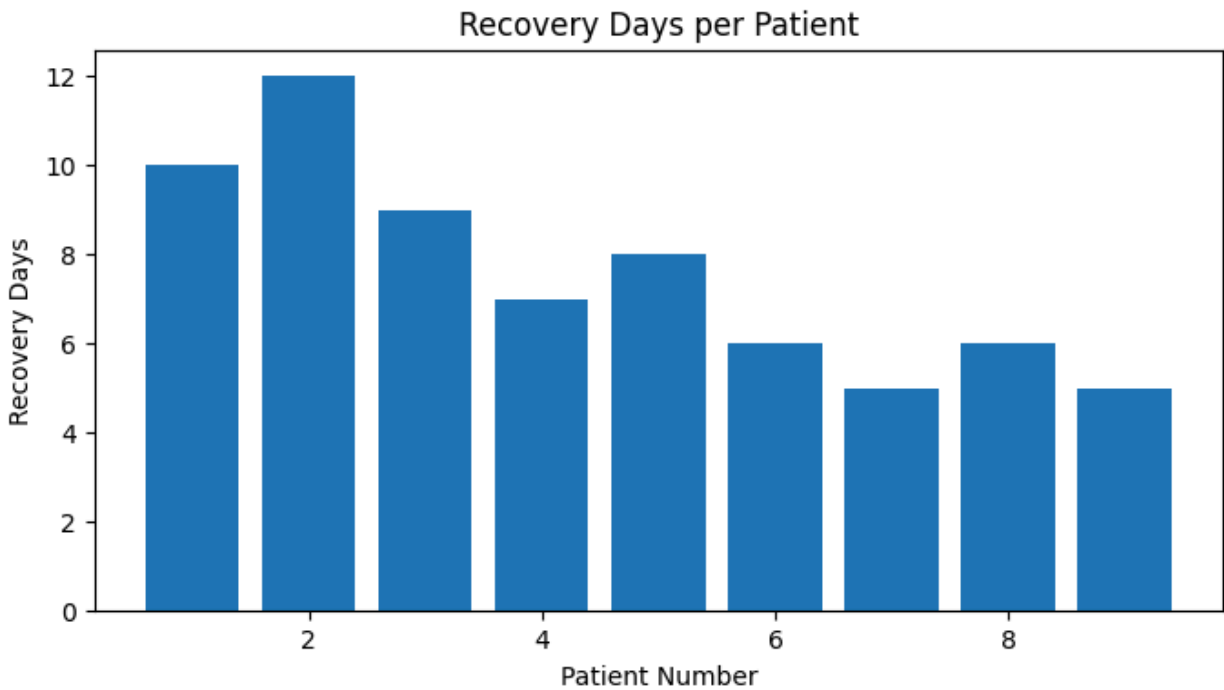
treatment	reduction
A	16.67
B	31.67
C	41.67



Recovery Days per Patient

Recovery duration for each patient record.

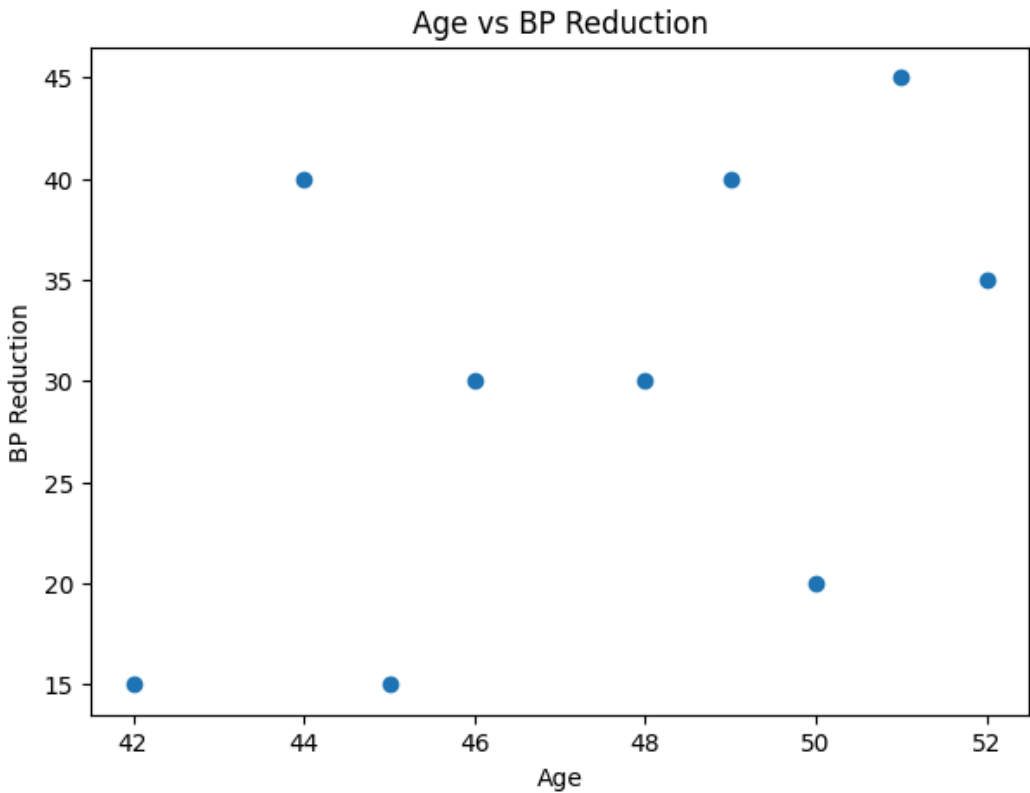
patient	days
P1	10
P2	12
P3	9
P4	7
P5	8
P6	6
P7	5
P8	6
P9	5



Age vs BP Reduction

Relationship between patient age and blood pressure reduction.

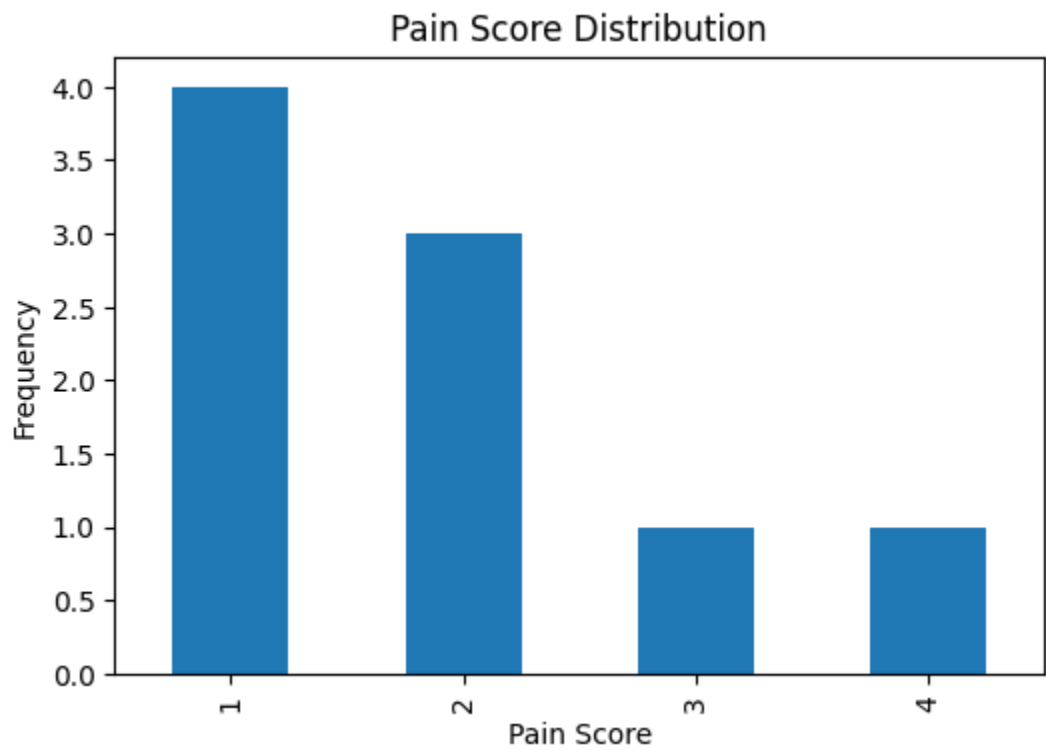
reductio n	ag e
15	45
20	50
15	42
30	48
35	52
30	46
40	44
40	49
45	51



Pain Score Distribution

Frequency of pain scores among patients.

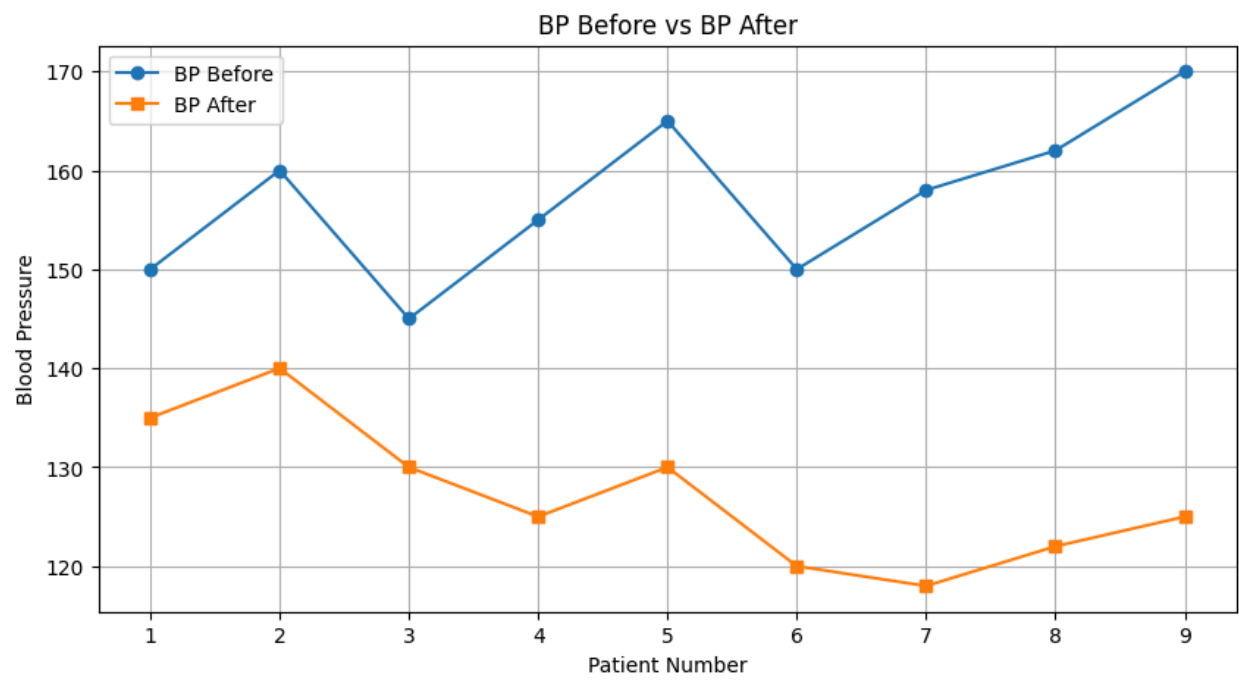
score	count
1	4
2	3
3	1
4	1



Blood Pressure Before and After Treatment

Comparison for all 9 patients.

patient	after	before
P1	135	150
P2	140	160
P3	130	145
P4	125	155
P5	130	165
P6	120	150
P7	118	158
P8	122	162
P9	125	170



Healthcare Dataset Summary (9 Patients)

Dataset Overview

Metric	Value
Total Patients	9
Treatments	A, B, C
Features	10
Recovery Rate	55.56%
Avg BP Before	157.22
Avg BP After	127.22
Avg BP Reduction	30
Avg Recovery Days	7.56

Key EDA Findings

Treatment A

- BP Reduction: 15, 20, 15
- Average Reduction: **16.67**
- Recovery Rate: **0%**
- Longest Recovery Time

Treatment B

- BP Reduction: 30, 35, 30
- Average Reduction: **31.67**
- Recovery Rate: **66.7%**

Treatment C

- BP Reduction: 40, 40, 45
- Average Reduction: **41.67**
- Recovery Rate: **100%**

- Fastest Recovery (5–6 days)

Business Insight

Treatment effectiveness:

Treatment C > Treatment B > Treatment A

Statistical Test Summary

1. Paired T-Test

BP Before vs BP After

Objective: Did therapy reduce blood pressure?

Result

BP after treatment is substantially lower than BP before treatment.

Conclusion

✅ Treatment significantly reduced BP.

2. Independent T-Test

Recovery Days by Gender

Objective: Does gender affect recovery?

Result

Male and female recovery times are similar.

Conclusion

❌ Gender does not appear to significantly influence recovery.

3. One-Way ANOVA

BP Reduction by Treatment

Objective: Compare Treatment A, B, C.

Result

Large difference in mean BP reduction:

- A = 16.67
- B = 31.67
- C = 41.67

Conclusion

✅ Treatment type significantly impacts BP reduction.

4. Chi-Square Test

Treatment vs Recovery

Objective: Check association between treatment and recovery.

Result

Recovery rate increases from A → B → C.

Conclusion

✅ Strong association between treatment type and recovery status.

5. Z-Test

Recovery Proportions

Comparison	Recovery Rate
A vs B	0% vs 66.7%
A vs C	0% vs 100%
B vs C	66.7% vs 100%

Conclusion

Treatment C consistently performs best.

Machine Learning Summary

Classification Models

Target = Recovered (Yes/No)

Logistic Regression

Use: Predict recovery probability.

Finding:

Recovery strongly influenced by:

- Treatment Type
- BP Reduction
- Recovery Days

Decision Tree

Use: Recovery prediction using rules.

Example Rule:

```
IF BP Reduction > 35
  Recovered = Yes
ELSE
  Recovered = No
```

Finding:

Easy interpretation for doctors.

Random Forest

Use: Multiple trees for better prediction.

Most Important Features

1. BP Reduction
2. Treatment
3. Recovery Days
4. Pain Score

Finding:

Best classification algorithm for this dataset.

KNN

Use: Similar patient prediction.

Example:

New patient

↓

Find nearest patients

↓

Predict recovery

Finding:

Patients with high BP reduction tend to recover.

Naive Bayes

Use: Probability-based prediction.

Finding:

Quick baseline model.

Regression Models

Linear Regression

Target = BP Reduction

Objective

Predict expected BP reduction.

Finding

Treatment type is strongly correlated with BP reduction.

Decision Tree Regressor**Finding**

Captures non-linear relationships better than linear regression.

Random Forest Regressor

Finding

Most robust regression model.

Can estimate:

- Recovery Days
 - BP Reduction
-

Clustering Algorithms

K-Means Clustering

Objective

Group similar patients.

Likely Clusters

Cluster	Type
Cluster 1	Low Improvement
Cluster 2	Medium Improvement
Cluster 3	High Improvement

Finding

Treatment C patients naturally form a high-performance cluster.

DBSCAN

Objective

Identify dense groups and outliers.

Finding

Can detect unusual patients.

Example:

Very high BP

Very slow recovery

would become an outlier.

Agglomerative Clustering

Objective

Build hierarchy of patient similarity.

Finding

Most suitable clustering algorithm for this small dataset.

Patients from Treatment C tend to merge into the same cluster.

PCA (Principal Component Analysis)

Objective

Reduce dimensions while preserving information.

Finding

Most variation explained by:

- 1. BP Reduction
- 2. BP Before
- 3. Recovery Days

Benefit

Visualize patients in 2D.

Feature Importance Summary

Most influential variables:

Rank	Feature
1	BP Reduction
2	Treatment
3	Recovery Days
4	BP After
5	Pain Score
6	Age
7	Gender

Final Healthcare Insights

Best Treatment

Treatment C

- Highest BP reduction
 - 100% recovery
 - Lowest pain score
 - Fastest recovery
-

Moderate Treatment

Treatment B

- Moderate BP reduction
 - Moderate recovery
-

Worst Treatment

Treatment A

- Lowest BP reduction
 - No recovered patients
 - Longest recovery duration
-